# WEEK-6

**AIM:**

Implementation of Paging Technique of Memory Management

**ALGORITHM:**

Step 1: Start

Step 2: Input

- Read the size of logical memory.
- Read the size of physical memory.
- Read the size of each page.
- Calculate the number of pages.
- Calculate the number of frames.

Step 3: Create Page Table

1. Create a page table.
2. Read the frame number allocated to each page.
3. Store the page-to-frame mapping in the page table.

Step 4: Read Logical Address

1. Read the logical address.
2. Calculate the page number:
    - Page Number = Logical Address / Page Size
3. Calculate the offset:
    - Offset = Logical Address % Page Size

Step 5: Address Translation

1. Use the page number to search the page table.
2. Find the corresponding frame number.
3. Calculate the physical address:
    - Physical Address = (Frame Number × Page Size) + Offset

Step 6: Output

- Display the page number.
- Display the offset.
- Display the frame number.
- Display the physical address.

Step 7: Stop

**PROGRAM:**

```
#include <stdio.h>

int main()
{
    int pageSize, logicalMemory, physicalMemory;
    int numPages, numFrames;
    int pageTable[100];
    int logicalAddress;
    int pageNumber, offset;
    int frameNumber, physicalAddress;
    int i;

    printf("Enter Logical Memory Size: ");
    scanf("%d", &logicalMemory);

    printf("Enter Physical Memory Size: ");
    scanf("%d", &physicalMemory);

    printf("Enter Page Size: ");
    scanf("%d", &pageSize);
```

```
numPages = logicalMemory / pageSize;
numFrames = physicalMemory / pageSize;

printf("\nNumber of Pages = %d", numPages);
printf("\nNumber of Frames = %d\n", numFrames);

printf("\nEnter Frame Number for each Page:\n");

for (i = 0; i < numPages; i++)
{
   printf("Page %d -> Frame: ", i);
   scanf("%d", &pageTable[i]);
}

// Display Page Table
printf("\n--- Page Table ---\n");
printf("Page\tFrame\n");

for (i = 0; i < numPages; i++)
{
   printf("%d\t%d\n", i, pageTable[i]);
}

printf("\nEnter Logical Address: ");
scanf("%d", &logicalAddress);


pageNumber = logicalAddress / pageSize;
```

```
    offset = logicalAddress % pageSize;

    if (pageNumber >= numPages)
    {

     printf("\nInvalid Logical Address!\n");
        return 0;
    }
   frameNumber = pageTable[pageNumber];
  physicalAddress = (frameNumber * pageSize) + offset;
  printf("\n--- Address Translation ---\n");
  printf("Logical Address  = %d\n", logicalAddress);
  printf("Page Number      = %d\n", pageNumber);
  printf("Offset           = %d\n", offset);
    printf("Frame Number     = %d\n", frameNumber);
    printf("Physical Address = %d\n", physicalAddress);
    return 0;
}
```

**OUTPUT:**

**TESTCASE 1:**

# VNR VJIET

**Name of the Laboratory:**

**Name of the Experiment:**

**Experiment No:** **Date:**

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS\"
Enter Logical Memory Size: 200
Enter Physical Memory Size: 400
Enter Page Size: 25

Number of Pages = 8
Number of Frames = 16

Enter Frame Number for each Page:
Page 0 -> Frame: 4
Page 1 -> Frame: 12
Page 2 -> Frame: 6
Page 3 -> Frame: 2
Page 4 -> Frame: 14
Page 5 -> Frame: 8
Page 6 -> Frame: 1
Page 7 -> Frame: 10

--- Page Table ---
Page    Frame
0       4
1       12
2       6
3       2
4       14
5       8
6       1
7       10

Enter Logical Address: 135

--- Address Translation ---
Logical Address = 135
Page Number = 5
Offset = 10
Frame Number = 8
Physical Address = 210
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

**TESTCASE2:**

```
cd "c:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc paging.c -o paging } ; if ($?) { .\paging }
Enter Logical Memory Size: 100
Enter Physical Memory Size: 200
Enter Page Size: 10

Number of Pages = 10
Number of Frames = 20

Enter Frame Number for each Page:
Page 0 -> Frame: 5
Page 1 -> Frame: 8
Page 2 -> Frame: 2
Page 3 -> Frame: 12
Page 4 -> Frame: 6
Page 5 -> Frame: 15
Page 6 -> Frame: 1
Page 7 -> Frame: 10
Page 8 -> Frame: 18
Page 9 -> Frame: 4

--- Page Table ---
Page    Frame
0       5
1       8
2       2
3       12
4       6
5       15
6       1
7       10
8       18
9       4

Enter Logical Address: 25

--- Address Translation ---
Logical Address = 25
Page Number = 2
Offset = 5
Frame Number = 2
Physical Address = 25
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

**TESTCASE 3:**

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc paging.c -o paging }
 ; if ($?) { .\paging }
Enter Logical Memory Size: 120
Enter Physical Memory Size: 240
Enter Page Size: 20

Number of Pages = 6
Number of Frames = 12

Enter Frame Number for each Page:
Page 0 -> Frame: 3
Page 1 -> Frame: 7
Page 2 -> Frame: 1
Page 3 -> Frame: 9
Page 4 -> Frame: 5
Page 5 -> Frame: 11

--- Page Table ---
Page    Frame
0       3
1       7
2       1
3       9
4       5
5       11

Enter Logical Address: 75

--- Address Translation ---
Logical Address = 75
Page Number = 3
Offset = 15
Frame Number = 9
Physical Address = 195
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```